What is the Relationship between Inflation and Unemployment in India: Is the Phillips Curve Still Valid?

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1. Introduction

The relationship between inflation and unemployment has been a key focus in studies of the broader economy, with the Phillips Curve as a basic model. This curve points to an opposite connection between inflation and unemployment—as prices go up, fewer people are out of work, and the other way around. However, experts have started to doubt this idea over time when looking at major changes in economic structure, government policy, and global developments.

In India, how prices and jobs affect each other depends on multiple factors. These include the Reserve Bank of India's monetary policy, supply chain issues, labor market functioning, and fiscal policy decisions. Sometimes when prices went up, fewer people were unemployed, but there have also been periods when both inflation and unemployment rose together. This contradicts the traditional Phillips Curve relationship.

This study investigates how inflation and unemployment are connected in India, assessing whether the Phillips Curve still holds in today's economic context. By analyzing historical trends, policy shifts, and statistical data, it aims to understand whether the Phillips Curve remains valid in India's evolving economy.

2. Research Question and Course Relevance

The research question is: *"What is the relationship between inflation and unemployment in India: Is the Phillips Curve still valid?"*

This question is closely related to the course **Macroeconomics-I**, which discusses the trade-offs between inflation and unemployment as part of macroeconomic theory. The Phillips Curve, central to this discussion, suggests a short-term inverse relationship between inflation and unemployment. Understanding this link helps evaluate policy trade-offs in developing economies such as India.

The course covers key macroeconomic indicators like inflation, unemployment, and fiscal-monetary policy. This project applies those concepts to the Indian economy to test whether the Phillips Curve theory aligns with recent data and trends.

3. Data Source

The data used in this study were obtained from the **World Bank Open Data** platform, which provides global development data compiled from official international sources. Specifically, the inflation rate (consumer prices) and unemployment rate of India were used.

4. Data Analysis

4.1 Unemployment Trends

Figure 1: India's Unemployment Rate (2000–2023)

Between 2000 and 2018, the unemployment rate in India remained relatively stable between 7.5% and 7.8%. However, a sharp decline occurred after 2018, followed by an increase during 2020, likely reflecting the economic impact of the COVID-19 pandemic and associated lockdowns.

4.2 Inflation Trends

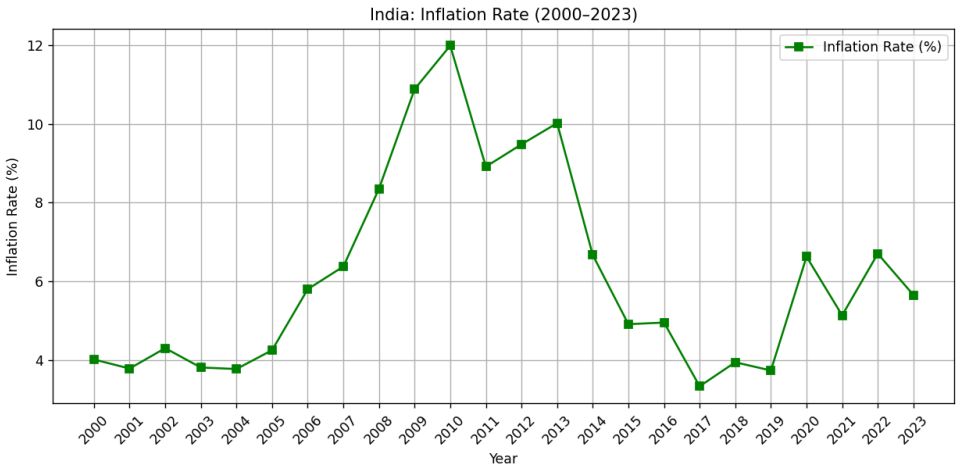

Figure 2: Inflation Rate in India (2000–2023)

The inflation rate in India has shown high volatility, reflecting domestic and global economic pressures. It peaked around 12% in 2010 due to global commodity price surges and supply-side issues, declined after 2013, and rose again around 2020 amid pandemic-related disruptions.

4.3 Phillips Curve Observation

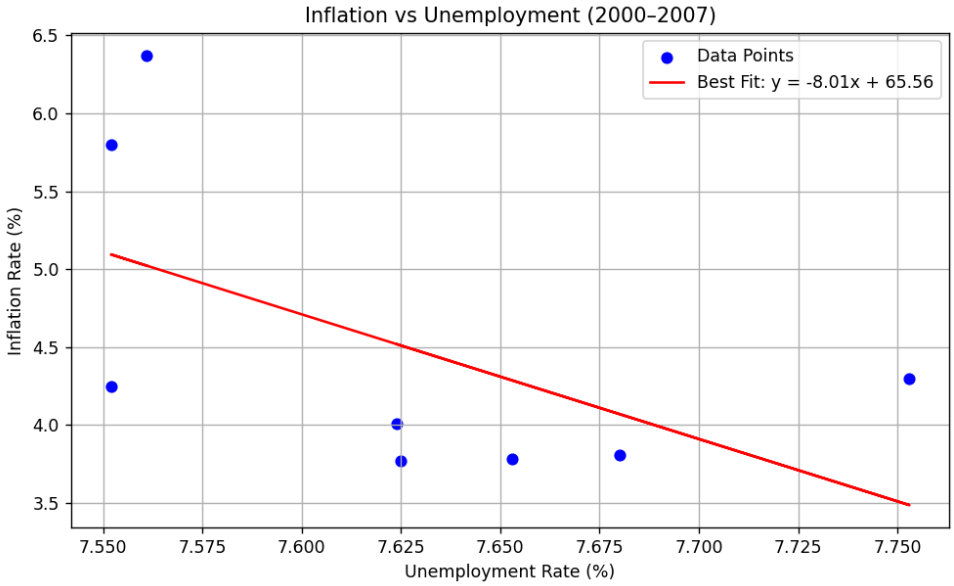

Figure 3: Inflation vs. Unemployment (2000–2007)

A clear negative relationship between inflation and unemployment was visible in early years (2000–2007), consistent with the Phillips Curve theory. However, when later years are included, the relationship becomes inconsistent. Periods where both inflation and unemployment moved together indicate the influence of external shocks and policy shifts.

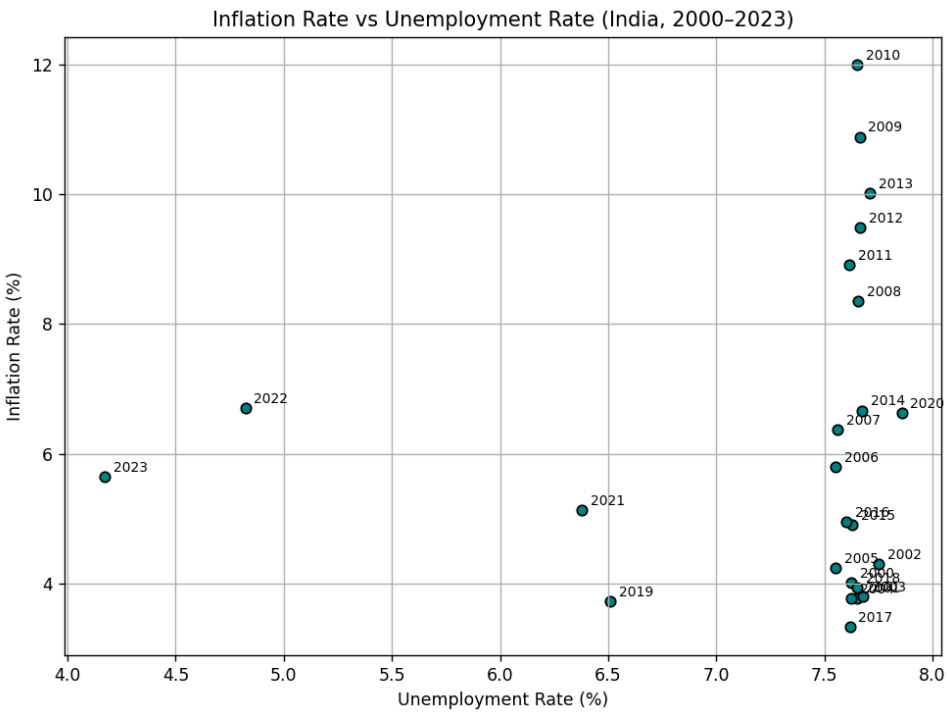

Figure 4: Inflation vs. Unemployment (2000–2023)

The long-run data shows no stable inverse relationship between inflation and unemployment. Instead, it supports the vertical long-run Phillips Curve concept, implying no systematic trade-off in the long term.

5. Findings and Discussion

The empirical analysis suggests that the Phillips Curve partially holds in the short run, showing a temporary trade-off between inflation and unemployment. However, this relationship breaks down in the long run. Periods of stagflation and inflation unrelated to employment confirm that other factors, such as supply shocks, inflationary expectations, and labor market structures, play a major role.

In the long run, the evidence supports the vertical Phillips Curve hypothesis—unemployment tends to revert to its natural rate, and inflation is mainly influenced by monetary factors rather than employment dynamics.

6. Conclusion

This research explores the inflation-unemployment relationship in India over 2000–2023, evaluating the validity of the Phillips Curve in short and long horizons. The findings confirm that while a short-run trade-off exists during certain periods, the relationship weakens in the long run. India's experience of stagflation and supply shocks demonstrates that structural reforms, efficient labor markets, and inflation targeting are crucial for sustainable growth and stability.

Source Data

`https://databank.worldbank.org/source/world-development-indicators`